

GARI, Russian Federation

WMO: 280490

Lat: 59.433N

Lon: 62.333E

Elev: 429

StdP: 14.47

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-22.4	-32.1	0.9	-26.7	-27.4	1.2	-22.3	22.6	8.3	20.6	10.3	6.8	0	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	84.0	67.5	80.9	65.9	77.5	64.0	69.8	80.6	67.8	78.3	65.8	74.7	7.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.1	75.2	63.9	90.3	72.7	62.0	84.3	71.0	34.0	80.7	32.4	78.4	30.8	74.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	19.1	17.1	-31.0	89.1	7.0	3.3	-36.1	91.5	-40.2	93.4	-44.1	95.3	-49.2	97.7		
(5)			-31.2	72.9	7.0	2.7	-36.2	74.9	-40.3	76.4	-44.2	77.9	-49.3	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	3.3	7.4	23.3	37.0	50.8	60.6	64.7	59.0	48.5	35.8	17.7	7.8		
	DBStd	23.93	13.29	13.96	9.50	9.55	9.88	8.64	8.08	7.68	7.93	9.24	13.70	14.55		
	HDD50	6854	1447	1193	829	398	115	14	3	9	120	450	968	1308		
	HDD65	11215	1912	1613	1294	839	451	185	108	219	496	907	1418	1773		
	CDD50	1308	0	0	0	9	139	331	458	287	76	8	0	0		
	CDD65	196	0	0	0	0	10	52	99	33	2	0	0	0		
	CDH74	1544	0	0	0	0	100	438	766	229	11	0	0	0		
	CDH80	355	0	0	0	0	9	100	183	63	0	0	0	0		
(14) Wind	WSAvg	8.6	8.8	8.6	9.6	9.1	9.3	8.7	7.4	7.6	8.1	8.5	8.6	8.5		
(15) Precipitation	PrecAvg	20.1	0.9	0.6	0.8	0.9	1.8	2.7	3.0	3.3	2.8	1.4	1.1	0.9		
	PrecMax	30.0	2.2	1.4	1.5	1.8	3.0	6.6	7.2	7.4	6.4	2.8	2.0	1.6		
	PrecMin	14.1	0.1	0.1	0.0	0.0	0.7	0.4	0.1	0.8	0.7	0.4	0.2	0.3		
	PrecStd	3.0	0.4	0.3	0.4	0.4	0.7	1.6	1.5	1.6	1.5	0.7	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	36.4	46.5	67.3	81.3	86.8	88.2	86.8	75.4	61.7	43.8	36.0		
		MCWB	33.0	32.6	37.7	51.5	59.4	67.8	69.6	69.9	60.9	51.8	41.0	33.5		
	2%	DB	28.9	33.2	41.8	60.2	77.0	82.8	84.5	79.9	69.5	57.2	39.8	33.1		
		MCWB	27.3	30.6	35.1	48.4	58.1	65.5	68.5	67.0	58.5	49.4	37.6	30.9		
	5%	DB	25.1	29.8	38.4	55.6	72.2	79.1	81.7	75.1	64.5	53.0	36.4	29.7		
		MCWB	24.0	27.6	33.0	45.2	55.5	64.2	67.3	64.3	55.9	47.1	34.2	28.1		
	10%	DB	21.2	26.0	35.6	51.4	67.2	75.1	78.5	71.2	60.1	48.9	33.4	26.2		
		MCWB	20.1	23.8	31.5	42.2	53.4	62.3	66.0	62.2	53.9	44.4	31.5	24.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	33.4	39.0	53.3	62.3	70.7	72.7	71.4	63.4	53.1	41.2	33.9		
		MCDB	34.0	35.7	44.3	63.4	75.2	82.1	82.1	82.0	71.8	59.6	43.1	35.5		
	2%	WB	27.5	30.9	36.0	49.4	59.2	67.9	70.5	68.1	59.3	50.3	37.8	31.1		
		MCDB	28.8	33.3	40.3	59.2	73.4	79.4	80.8	77.4	67.3	55.7	39.6	32.9		
	5%	WB	24.2	27.8	33.6	45.6	56.9	65.5	68.8	65.5	56.9	47.6	34.6	28.2		
		MCDB	25.1	29.8	37.8	54.1	69.4	75.9	79.4	73.3	63.1	52.8	36.1	29.7		
	10%	WB	20.2	24.0	31.7	43.0	54.5	63.4	67.0	63.3	54.5	44.7	31.7	24.8		
		MCDB	21.2	25.9	35.0	50.3	65.5	73.5	76.9	69.9	59.4	48.6	33.1	26.0		
(35) Mean Daily Temperature Range	MDBR		10.9	12.9	13.8	15.0	17.3	16.4	16.1	14.4	12.9	10.3	9.8	10.5		
	5% DB	MCDBR	12.6	11.8	15.3	20.6	23.4	21.0	19.3	19.5	18.7	15.0	9.8	10.8		
		MCWBR	12.1	9.9	11.0	13.0	11.6	10.0	8.5	9.5	10.0	9.3	8.4	10.1		
	5% WB	MCDBR	12.7	11.6	13.1	19.4	21.3	19.5	18.6	17.4	16.7	14.0	9.1	10.8		
		MCWBR	12.4	10.1	10.1	12.9	11.9	10.2	9.5	9.6	9.8	9.4	8.1	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.302	0.355	0.393	0.393	0.394	0.408	0.418	0.367	0.331	0.285	0.256		
	taud		2.018	2.043	1.985	2.100	2.295	2.350	2.308	2.250	2.358	2.315	2.126	1.967		
	Ebn at Noon		156	215	237	253	265	266	258	244	239	209	150	112		
	Edh at Noon		16	27	39	42	37	36	36	36	27	21	14	11		
(44) All-Sky Solar Radiation	RadAvg		154	443	884	1299	1614	1715	1708	1248	749	383	158	81		
	RadStd		13	37	70	128	136	129	149	126	80	54	23	11		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page